

SAMBP Research Gap Analysis & TR Roadmap

Physics-Informed Protection for IBR-Dominated Microgrids

Author Anoop V Eluvathingal
Lab Smart Grid Computational Research Lab (SGCRL), IIT Madras
Guide Prof. K. Shanti Swarup, Dept. of Electrical Engineering
Date April 2026
Version 1.0 — Internal Draft
Purpose Hiring conversations · Consortium scoping · TR allocation

Confidential. This document is an internal working draft intended for hiring and consortium scoping discussions. It contains honest scope boundaries and effort estimates that should not be shared publicly without review.

Contents

1	Purpose and Context	1
1.1	Current Framework in Brief	1
2	Gap Classification Methodology	1
3	Full Gap Table	2

4	Effort & Priority Summary	4
4.1	Aggregated Effort by Tier	4
4.2	Recommended Sequencing	5
5	Hiring Profiles	5
5.1	Hire #1 — BESS / Power Electronics Protection (TR-68, TR-74, TR-85)	5
5.2	Hire #2 — Wind (Type-4) & Subsynchronous Interactions (TR-69, TR-75)	5
5.3	Hire #3 — Network Topology & Wide-Area Protection (TR-70, TR-83)	6
5.4	Hire #4 — Cybersecurity & Communications (TR-82, TR-76)	6
6	Consortium Workpackage Mapping	6
7	What Is Not a Gap (Scope Bounds)	7
8	Document Maintenance	8

1. Purpose and Context

This document records the output of a systematic gap analysis of the SAMBP (Substation Adaptive Microgrid Backup Protection) simulation framework, benchmarked against the coverage a Horizon Europe or CIGRE WG B5 consortium would expect from an IBR protection research programme.

It serves three audiences:

- 1. **Hiring:** defines the technical scope for the next research engineer or postdoc — each Tier-1 gap is a self-contained project.
- 2. **Consortium scoping:** identifies which workpackage slots map to which TRs, and which gaps represent genuine collaboration value-adds vs. in-house extensions.
- 3. **Thesis planning:** provides a critical-path view of what must be done before Q3 2028 submission and what can be flagged as future work.

1.1. Current Framework in Brief

SAMBP currently delivers four complete relay families, each following a uniform five-layer architecture:

Relay	Element	Key Result	Status
87L	Line current differential	12/12 scenarios, 15 k MC	Done
87T	Transformer differential	7/7 events, CT-sat veto	Done
87B	Bus differential	6/6 events, $\kappa_n < 6.3$	Done
OC	Adaptive overcurrent	N-gen greedy cascade	Done
21/21P	Mho distance (pilot)	5 loc $\times$ 3 types	Active
40/64G/78/81	Generator suite	LOE, sub-harm, GFM out-of-step	Active

Source types currently covered: synchronous generator (SG), DFIG (Type-3 wind), GFL-PV, GFM inverter. Network topology: radial only (5-node + IEEE 39-bus reference).

2. Gap Classification Methodology

Gaps are classified into three tiers:

- **T1 Critical** — a consortium partner will identify this as a missing deliverable; absence weakens the overall thesis claim.
- **T2 Significant** — raised in technical peer review; should be addressed or explicitly bounded as future work with justification.
- **T3 Emerging** — forward-looking; relevant for 2027+ grid standards and positions the lab for follow-on funding.

Effort estimates are in **engineer-months** (EM), calibrated to a postdoc/senior PhD student working on SAMBP code and documentation. One EM assumes $\sim$ 160 hours of focused research+coding time.

3. Full Gap Table

The table below is the primary deliverable of this document. Columns: Gap, Tier, Proposed TR, Effort (EM), Dependencies, and notes on hiring/consortium fit.

Gap	Tier	TR	Effort	Depends on	Hiring / Consortium note
Tier 1 — Critical: Consortium will ask for these explicitly					
BESS source model GFM-BESS, VSM mode fault current; near-zero natural fault current; virtual impedance control	T1	TR-68	3–4 EM	<i>TR-05 (87L), TR-07 (87B); BESS model lib</i>	Ideal first hire project. Self-contained: write BESS current model, extend 87L/87B, run 1 k MC. Outcome: TR-68 + journal paper.
Type-4 / PMSG wind Full back-to-back converter; negligible neg-seq injection; no crowbar; inrush blocking must be revisited	T1	TR-69	3 EM	<i>TR-56 (DFIG); ANDES PMSG model</i>	Strong fit for a hire with converter controls background. Pairs with TR-56 ANDES integration already active.
Meshed / ring topology Directional relaying (67), pilot schemes, loop-flow; all current models are radial	T1	TR-70	5–6 EM	<i>TR-66 (IEEE 39-bus); network layer rewrite</i>	Largest structural gap. Requires new network layer in PandaPower; directional OC and pilot logic are new relay families. Consortium WP-level item.
Auto-reclose (79) on IBR feeders IEEE 1547-2018 ride-through; delayed reclose; live-line / dead-line voltage supervision	T1	TR-71	2–3 EM	<i>TR-05 (87L); IEEE 1547 ride-through study</i>	Bounded scope; good second hire project. Standalone relay-79 FSM. Direct CIGRE WG B5 deliverable.
Evolving & cross-country faults SLG → 3-phase escalation; simultaneous faults at two locations; IBR current-limiting complicates detection	T1	TR-72	2 EM	<i>TR-05 (87L); fault scenario engine</i>	Extends existing fault scenario library. Low implementation risk; high consortium visibility.
Tier 2 — Significant: Should be addressed or explicitly bounded					
Negative-seq protection under GFL dominance 46/47 relay blind spot when GFL is dominant source; neg-seq too small for conventional pickup	T2	TR-73	1.5 EM	<i>TR-05, TR-07; GFL dominance scenario</i>	In-house extension. Sequence estimator already in <code>sync_oc</code> ; extend to 46/47 threshold study.

Gap	Tier	TR	Effort	Depends on	Hiring / Consortium note
IEEE 1547-2018 ride-through compliance Mandatory ride-through window (momentary cessation vs. continuous); protection must not mal-operate	T2	TR-74	2 EM	TR-71 (79 relay); GFL/GFM models	Pairs naturally with TR-71. Consortium standard compliance deliverable.
SSCI / Subsynchronous resonance DFIG + series-compensated line interaction (5–40 Hz); protection must distinguish resonance from fault	T2	TR-75	3 EM	TR-56 (DFIG); series-comp line model	High academic novelty. ERCOT 2009 case study provides validation baseline. New ANDES model needed.
COMTRADE / field-data validation All validation is synthetic or ANDES-simulated; COMTRADE import pipeline absent	T2	TR-76	1.5–2 EM	Data access agreement with utility / EPRI	Single highest-impact credibility move. Requires utility data-sharing MOU. Can be funded as industry collaboration.
Open-conductor (series) faults Broken conductor, blown fuse; unbalanced supply; dangerous for IBR 3-phase controllers	T2	TR-77	1.5 EM	TR-05 (87L); sequence tools	Adds 46/open-phase element. Medium effort; strong safety narrative for consortium.
Sympathetic tripping — parallel feeders Transformer inrush on healthy feeder during fault on adjacent feeder can mal-operate 87T	T2	TR-78	1 EM	TR-03 (87T); parallel feeder model	Short study; extends existing 87T event library with sympathetic-inrush event.
N-2 / degraded relay set Performance under single-relay failure, CT bypass, or comms loss beyond 87L FSM fallback	T2	TR-79	1.5 EM	TR-05, TR-07; fallback logic	Directly supports reliability quantification. Extends existing <code>line_fallback_logic.py</code> .
Tier 3 — Emerging: Positions lab for 2027+ funding & standards					
DC microgrids & hybrid AC/DC di/dt discrimination; DC CB coordination; data-centre, EV, marine applications	T3	TR-80	5 EM	New discipline; no current code base	Scope for a dedicated postdoc hire. Horizon Europe MVDC workpackage candidate.

Gap	Tier	TR	Effort	Depends on	Hiring / Consortium note
Geomagnetic disturbance (GMD/GIC) Quasi-DC GIC causes transformer half-cycle saturation → harmonic currents → 87T mal-op; NERC TPL-007	T3	TR-81	2.5 EM	TR-03 (87T); GIC transformer model	High regulatory visibility (NERC TPL-007). Transformer half-cycle saturation model needed in ANDES.
Cyber-physical attacks on GOOSE GPS spoofing, GOOSE replay, latency injection on confidence gate; IEC 62351 compliance	T3	TR-82	2.5 EM	TR-65 (GOOSE); IEC 62351 framework	Strong fit for cybersecurity-background hire. Pairs with ch6 security audit code already in codebase.
PMU / synchrophasor-based wide-area backup C37.118 measurement chain; wide-area backup using PMU data; latency-aware confidence gate	T3	TR-83	4 EM	TR-65–67 (system integration); PMU data access	Consortium WP item for grid-scale IBR projects. Requires field PMU data or real-time simulator.
Ferroresonance VT / transformer + cable capacitance; sustained oscillation; IBR-islanded systems are susceptible	T3	TR-84	2 EM	TR-03 (87T); non-linear core model	Extends transformer event library with 8th event. Academic novelty; targeted CIGRE paper.
Cold load pickup & motor starting Post-islanding load surge (6–8× rated); protection must not trip; BESS response critical	T3	TR-85	1.5 EM	TR-68 (BESS); islanding study	Small but high operational value. Extends <code>run_islanding_study.py</code> .

4. Effort & Priority Summary

4.1. Aggregated Effort by Tier

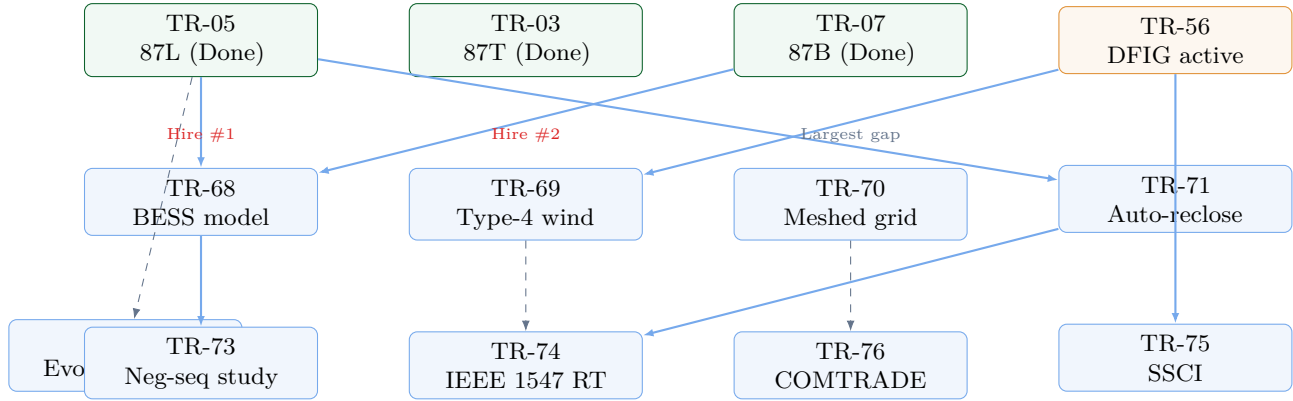
Tier	Gaps	TRs	Effort range (EM)	FTE-years
T1 Critical	5	TR-68–72	15–18 EM	1.5 FTE-years
T2 Significant	7	TR-73–79	12–15 EM	1.2 FTE-years
T3 Emerging	5	TR-80–85	17 EM	1.4 FTE-years
Total	17	TR-68–85	44–50 EM	≈4 FTE-years

Thesis scope note. TR-68 through TR-72 (Tier 1) are candidates for inclusion as Chapter 8 extensions before the Q3 2028 submission deadline, subject to hiring. TR-73 through TR-85 are

scoped as post-thesis consortium deliverables or future-work bounds in Chapter 10 (Conclusions).

4.2. Recommended Sequencing

The dependency graph below shows the recommended order of attack. Bold arrows are blocking dependencies; dashed arrows are soft couplings.



5. Hiring Profiles

5.1. Hire #1 — BESS / Power Electronics Protection (TR-68, TR-74, TR-85)

Title: Research Engineer / Postdoc — BESS & IBR Protection

Duration: 18–24 months

Deliverables: TR-68 (BESS source model + 87L/87B extension), TR-74 (IEEE 1547 ride-through compliance study), TR-85 (cold load pickup). Journal target: *IEEE Transactions on Power Delivery*.

Required background:

- Grid-scale BESS control (VSM / droop / virtual impedance modes)
- Python (NumPy/SciPy); familiarity with PandaPower or ANDES
- Protection fundamentals (IEC 60255, percentage-differential relays)

Nice to have: Real-time simulator experience (OPAL-RT, RTDS).

5.2. Hire #2 — Wind (Type-4) & Subsynchronous Interactions (TR-69, TR-75)

Title: Research Engineer — Wind IBR Protection

Duration: 12–18 months

Deliverables: TR-69 (PMSG Type-4 source model + protection study), TR-75 (SSCI detection and relay discrimination). Journal target: *IEEE Transactions on Energy Conversion*.

Required background:

- PMSG / Type-4 converter controls; crowbar-less FRT

- ANDES or PSCAD; understanding of subsynchronous resonance
- Relay sequence-component analysis (Fortescue)

5.3. Hire #3 — Network Topology & Wide-Area Protection (TR-70, TR-83)

Title: Research Engineer — Meshed Network & Synchrophasor Protection

Duration: 24 months (largest scope)

Deliverables: TR-70 (meshed topology, directional 67 relay, pilot scheme), TR-83 (PMU / C37.118 wide-area backup). Journal target: *IEEE Transactions on Smart Grid*.

Required background:

- Power system network analysis (meshed load flow, fault analysis in loops)
- Directional overcurrent and distance relay coordination
- Synchrophasor (PMU) measurement chains; preferably C37.118

5.4. Hire #4 — Cybersecurity & Communications (TR-82, TR-76)

Title: Research Engineer — Cyber-Physical Security

Duration: 12–18 months

Deliverables: TR-82 (GOOSE attack modelling, IEC 62351 compliance), TR-76 (COMTRADE data pipeline, field-data validation MOU). Journal target: *IEEE Transactions on Industrial Informatics*.

Required background:

- IEC 61850 GOOSE / MMS protocol internals
- Cyber-physical attack modelling (replay, spoofing, latency injection)
- Familiarity with IEC 62351 security for protection

6. Consortium Workpackage Mapping

The table below maps gaps to Horizon Europe / CIGRE workpackage structures. It is intended as a starting point for a consortium proposal — each WP corresponds to one or more hires and a coherent set of TRs.

WP	Title	TRs	Effort	Partner fit
WP1	BESS & GFM Protection	TR-68, 74, 85	6 EM	Industry (BESS OEM)
WP2	Type-4 Wind & SSCI	TR-69, 75	6 EM	Wind OEM / TSO
WP3	Meshed Grid & Directional	TR-70, 83	10 EM	TSO / DSO
WP4	Standards Compliance	TR-71, 72, 73, 77, 78, 79	10 EM	Relay vendor
WP5	Field Validation	TR-76	2 EM	Utility (data MOU)
WP6	Cyber-Physical Security	TR-82	3 EM	Cybersecurity partner
WP7	Emerging Topics	TR-80, 81, 84	10 EM	Academic partners
Total			47 EM	

Critical path to Q3 2028 submission. TR-68 (BESS) is the single most important new TR before submission. It closes the largest source-type gap and produces a chapter-length study. Hire #1 must be onboarded by **Q4 2026** to leave sufficient time for TR-68 + journal submission before the thesis lock date.

TR-70 (meshed grid) is the largest structural gap but is *not* on the critical path — it is scoped as post-thesis consortium work and explicitly bounded in Chapter 10.

7. What Is Not a Gap (Scope Bounds)

The following items were considered and deliberately excluded:

- **HVDC protection:** Out of scope for a microgrid/distribution protection thesis; a separate HVDC protection programme is warranted.
- **Arc-flash detection:** Overlap with occupational safety standards (NFPA 70E / IEC 62271); not a relay protection problem in the SAMPB sense.
- **Stator ground fault (64S) for synchronous machines:** Covered adequately by existing 64G work; marginal additional value.
- **Load shedding (UFLS) schemes:** System-stability function, not a selective protection function; separate discipline.
- **PSCAD as primary simulator:** ANDES + PandaPower + OpenDSS cover the required fidelity range. PSCAD is used in Ch. 6 data generation only; no additional PSCAD integration is planned.

8. Document Maintenance

This document should be updated:

- When a TR changes status (Planned → Active → Done).
- When a new hire is confirmed — assign TRs and update effort column.
- When a consortium proposal is submitted — attach WP mapping as appendix.
- Quarterly review cadence recommended; owner: Anoop V Eluvathingal.

Smart Grid Computational Research Lab (SGCRL), IIT Madras · Prof. K. Shanti Swarup · Internal Draft
v1.0 · April 2026 · Not for external distribution without review.